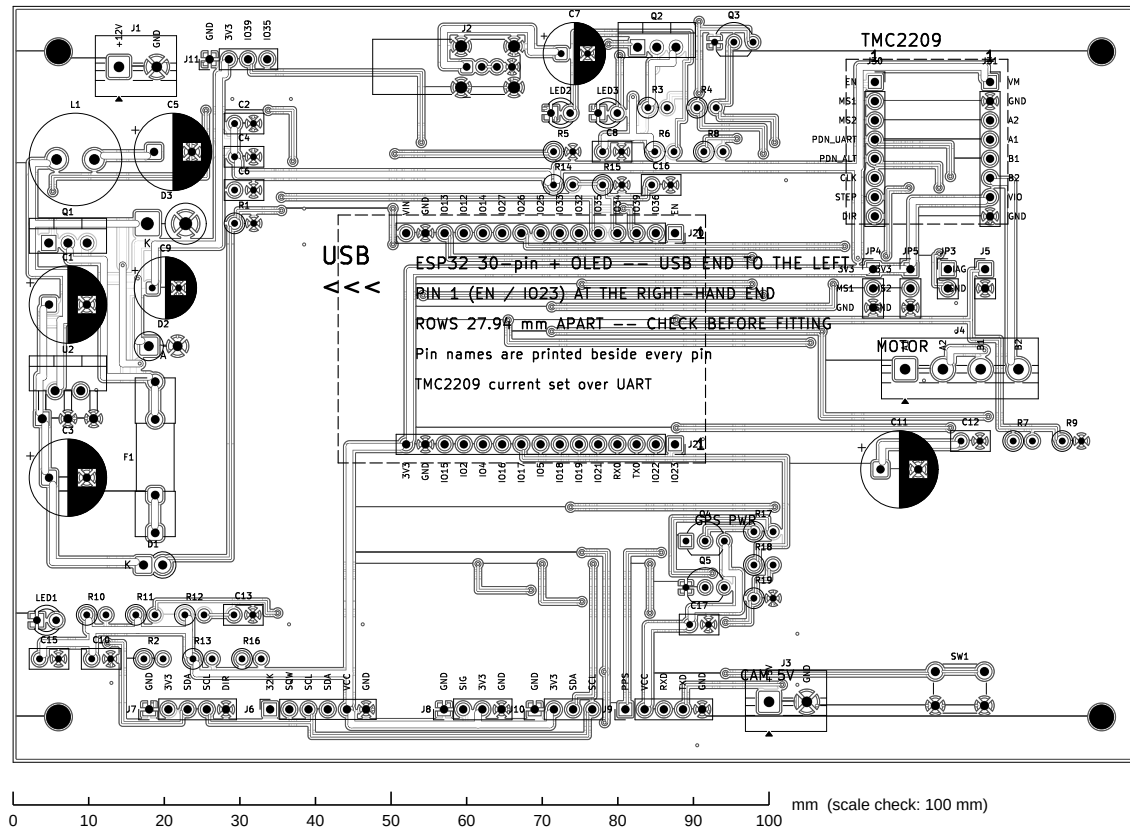


Star tracker controller Rev C - 1:1 fit template (top view, 150 x 100 mm)

Print at 100% / "Actual size" (no fit-to-page). Verify the scale bar below with a ruler. Dark = top copper, light grey = bottom copper (seen through), ring = drill hole.



Inventory check - BOM (tick when in hand)

Qty	Refs	Value	Footprint	Note
1	C1	470uF/25V	CP_Radial_D10.0mm_P5.00mm	
1	C11	100uF/25V LowESR	CP_Radial_D10.0mm_P5.00mm	VMOT bulk - MUST sit at the driver socket
10	C10 C12 C13 C15 C16 C17 C2 C4 C6 C8	100nF	C_Disc_D5.0mm_W2.5mm_P2.50mm	
1	C3	470uF/25V LowESR	CP_Radial_D10.0mm_P5.00mm	
1	C5	220uF/25V LowESR	CP_Radial_D10.0mm_P5.00mm	
1	C7	100uF/16V	CP_Radial_D8.0mm_P3.50mm	
1	C9	100uF/10V	CP_Radial_D8.0mm_P3.50mm	
1	D1	1N4744A 15V	D_DO-41_SOD81_P2.54mm_Vertical_CathodeUp	Vgs clamp, cathode to source
1	D2	P6KE20CA	D_DO-15_P3.81mm_Vertical_AnodeUp	
1	D3	1N5822	D_DO-201AD_P5.08mm_Vertical_CathodeUp	catch diode, cathode to SW node
1	F1	3A	Fuseholder_Clip-5x20mm_Littelfuse_111_Inline_P20.00x5.00mm_D1.05mm_Horizontal	
1	J1	12V IN	TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm	12-14.6V LiFePO4 in
1	J10	I2C EXP	PinHeader_1x04_P2.54mm_Vertical	v2 compass
1	J11	SPARE IO	PinHeader_1x04_P2.54mm_Vertical	the upper row's two free pins; both input-only
1	J2	CAM USB-A	USB_A_Wuerth_614004134726_Horizontal	D+/D- shorted = dedicated charging port
1	J20	ESP32 L	PinSocket_1x15_P2.54mm_Vertical	
1	J21	ESP32 R	PinSocket_1x15_P2.54mm_Vertical	
1	J3	CAM 5V	TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm	
1	J30	TMC2209 L	PinSocket_1x08_P2.54mm_Vertical	
1	J31	TMC2209 R	PinSocket_1x08_P2.54mm_Vertical	
1	J4	MOTOR	TerminalBlock_MaiXu_MX126-5.0-04P_1x04_P5.00mm	pins 1-2 one coil, 3-4 the other
1	J5	DIAG	PinHeader_1x02_P2.54mm_Vertical	
1	J6	DS3231	PinSocket_1x06_P2.54mm_Vertical	ZS-042 order: 32K SQW SCL SDA VCC GND
1	J7	AS5600	PinHeader_1x05_P2.54mm_Vertical	remote on the encoder bracket; pin5 = DIR->GND
1	J8	HOME	PinHeader_1x04_P2.54mm_Vertical	optional opto flag; GND on BOTH end pins so the pour can always reach one of them
1	J9	GPS	PinHeader_1x05_P2.54mm_Vertical	GY-NEO6MV2: pins 2-5 match its VCC/RX/TX/GND cable; PPS is a flying lead from the module's PPS LED pad
1	JP3	PDN ALT	PinHeader_1x02_P2.54mm_Vertical	
1	JP4	MS1	PinHeader_1x03_P2.54mm_Vertical	
1	JP5	MS2	PinHeader_1x03_P2.54mm_Vertical	
1	L1	33uH 3A	L_Radial_D12.0mm_P5.00mm_Fastron_11P	
1	LED1	GRN 5V	LED_D3.0mm	

Inventory check - BOM (cont.)

Qty	Refs	Value	Footprint	Note
1	LED2	GRN CAM	LED_D3.0mm	
1	LED3	RED ST	LED_D3.0mm	
2	Q1 Q2	IRF4905	TO-220-3_Vertical	reverse-polarity; drain to source of supply
2	Q3 Q5	2N3904	TO-92_Inline_Wide	
1	Q4	2N3906	TO-92_Inline_Wide	PNP high-side switch; the module's own LDO makes 3.3V from this
7	R1 R13 R16 R18 R3 R4 R9	10k	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	
2	R10 R11	4.7k	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	omit if the DS3231 module already has pull-ups (most do)
1	R12	4.7k	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	the wired-OR pull-up; always populate
1	R14	100k 1%	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	
1	R15	18k 1%	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	14.6V -> 2.23V at GPIO34
4	R17 R2 R6 R7	1k	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	
2	R19 R5	100k	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	holds camera off while GPIO floats
1	R8	330R	R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical	
1	SW1	CONFIG	SW_PUSH_6mm	also wakes from deep sleep via the GPIO4 wired-OR
1	U2	LM2596T-5.0	TO-220-5_P3.4x3.7mm_StaggerO dd_Lead3.8mm_Vertical	TO-220-5; needs a clip heatsink above ~1.5A